

Targeting Higher Credit Card Payments

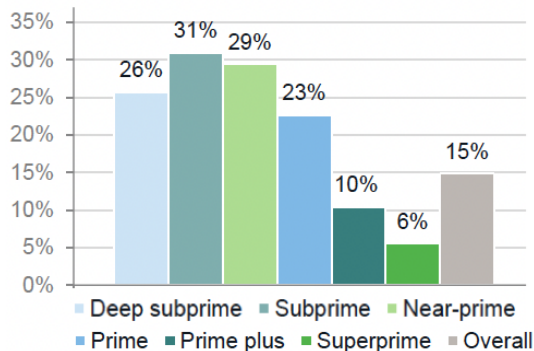
Benedict Guttman-Kenney

Rice University

Montana Financial Decision-Making Conference, 7th August 2026

Credit Cardholders Often Only Pay The Minimum Payment Amount. Why?

% of credit card accounts paying the minimum payment amount, by credit score
(Consumer Financial Protection Bureau, 2025)



29% of accounts pay within \$50 of the minimum payment amount
(Keys & Wang, 2019 JFE)

One Potential Explanation: Anchoring Heuristic

Anchoring and adjustment heuristic: Foundational bias in Tversky & Kahneman (1974)

Key idea:

- Anchors: Salient but irrelevant numbers can bias consumer choices
e.g., randomly-assigned last two-digits of Social Security Number (Ariely et al., 2003)
- Consumers choose at / near the numerical amount initially presented
& adjust away too little

“One of the most robust phenomena studied across the social behavioral sciences”
– 50-year meta-analysis by Schley and Weingarten (2025)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes “**anchoring**” as an important determinant of consumer credit card payment choices:

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Sunstein & Thaler, 2008)

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Key implication: Without anchoring, cardholders would ↑ credit card payments

This Paper: Do Online Experimental Findings Extrapolate To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech lender

- RCT that varies in-app (manual) credit card payment options for six statement cycles
 - Test shrouding minimum payment amount: Under anchoring, should $\uparrow$ payments
 - Test effects of 50% payment amount option
- Observe outcomes over six statement cycles

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Key Findings: Results of hypothetical online experiments do not translate into the field

- No increase in payments from shrouding the minimum payment
- Shrouding *lowers* initial payment choices: Opposite of what anchoring predicts
- 50% payment option also does not change payments

- Evidence that a **policy** shrouding the minimum payment is expected to be ineffective (Financial Conduct Authority, 2021; Brookings, 2022; Financial Health Network, 2023)

Contributions

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- **Household Finance:** Understand why low-credit score credit cardholders often only pay at or just-above the minimum payment amount
 - *Not* driven by heuristic anchoring to the minimum payment
 - Cardholders frequently lack liquid cash, not at borrowing constraint

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- **Behavioral Economics:** Weakness of active choice nudges & anchoring heuristic
- **Empirical methods** in social sciences & challenges of scaling interventions (List, 2022)
 - Online experiments are increasingly used across social sciences (Rauf et al., 2025)
 - Magnitudes can differ in online vs. field experiments (List, 2001; Gandhi et al., 2025)
 - Example where the direction of the key finding differs

1. The Experiment (Design, Data, Methodology)
2. Experimental Results
3. Mechanisms

1. The Experiment

Field Experiment Design

Control (C)

Make Payment - Current Balance (Default)

9:41

< Choose Amount

👉 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👉 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$25.00 Minimum Amount

Next

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Treatment T2

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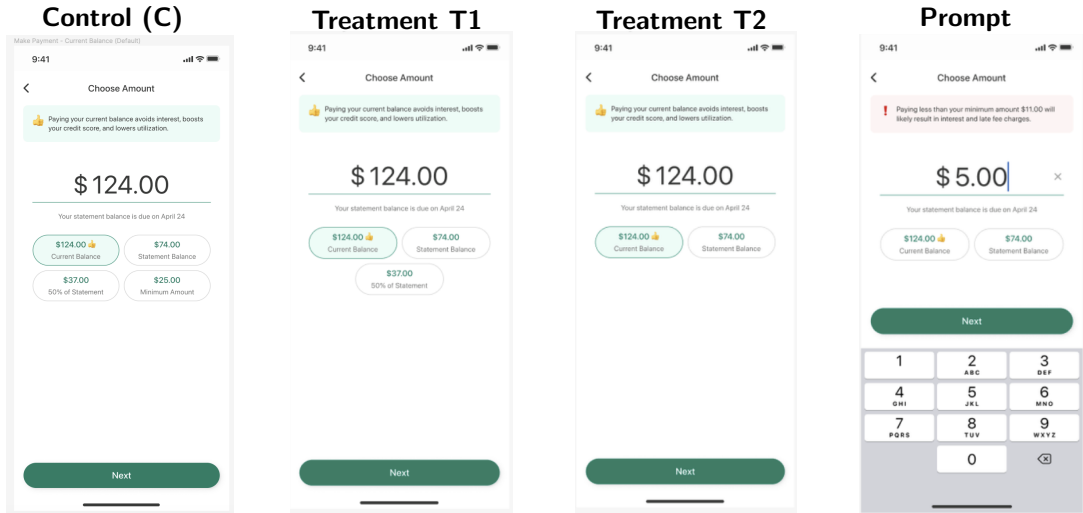
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--------------------------------------	-------------------------------------

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Field Experiment Design



If attempt to pay < minimum, a **prompt** reveals minimum payment

Field Experiment Design ($N = 6,714$)

Control (C)

4,542

Make Payment - Current Balance (Default)

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T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

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T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

T1 (vs T2): Effect of *adding* 50% option (conditional on shrouding minimum)

Arro is a FinTech lender that provides credit cards in the United States:

- Means: \$290 credit limits, \$200 statement balance, 560 VantageScore
- Clearly not representative of broader credit card market!

Below-prime credit score (< 660 VantageScore) consumers:

- Highest interest rates and so the large potential gains from increasing payments (Drechsler et al., 2025; Consumer Financial Protection Bureau, 2025)
- Most likely to pay at or just-above the minimum payment amount and have similar “anchoring” estimates to prime consumers (Keys & Wang, 2019)

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Datasets

Lender's administrative credit card data linked with: (1) Clickstream data on payment choices; (2) Plaid financial transaction data on liquid cash; (3) Equifax credit reporting data

Today, focus on regression results from dynamic specification:

$$Y_{i,t} = \sum_{\tau=0}^6 \left[\delta_{\tau}^{T1(vsC)} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2(vsC)} (D_{\tau} \times T2_i) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (1)$$

- X_i vector of pre-experiment controls, μ_t statement cycle fixed effects
- Standard errors clustered at consumer-level

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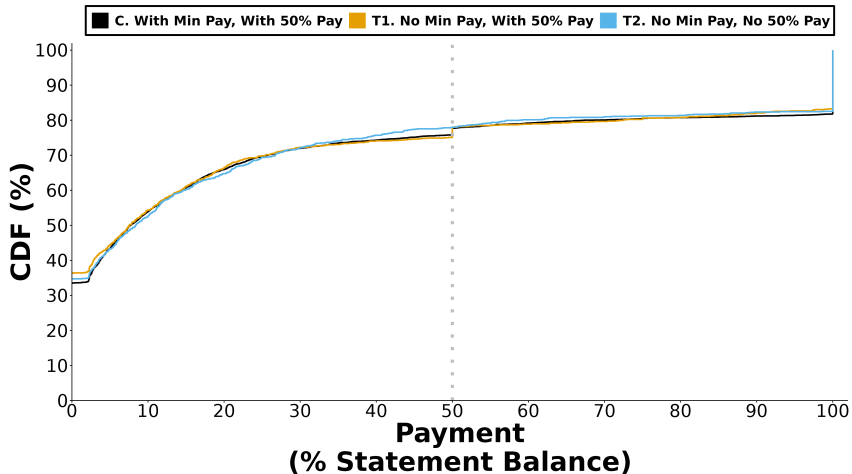
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Pooled specification for more statistical power:

$$Y_{i,t} = \delta^{T1(vsC)} T1_i + \delta^{T2(vsC)} T2_i + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (2)$$

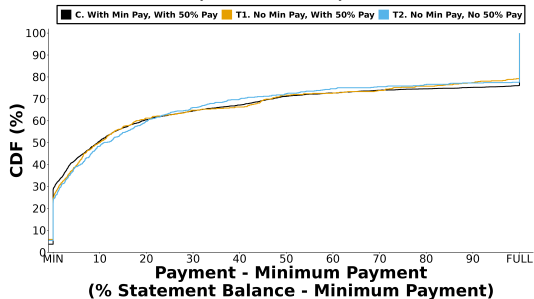
2. Experimental Results

Payment Distribution Shows Little Overall Change After 6 Cycles

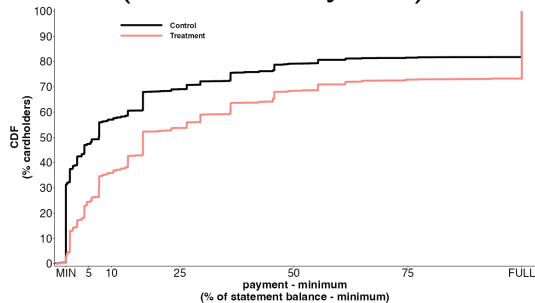


Field Experiment Results In Sharp Contrast To Online Experiments

Field Experiment (This Paper)



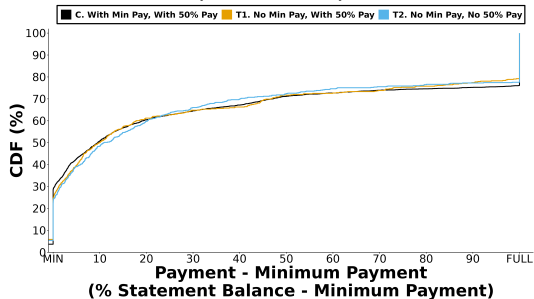
Online Experiment Example (Guttman-Kenney, 2024)



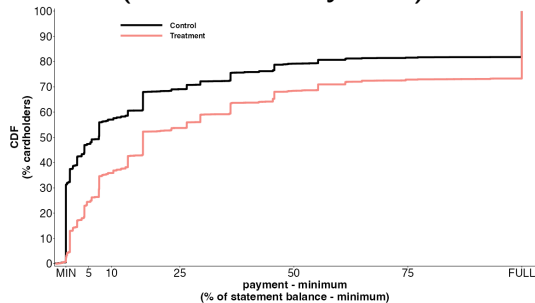
↑ 12.4*** p.p. (s.e. 0.8)
($N = 7,938$, mean 28.1%)

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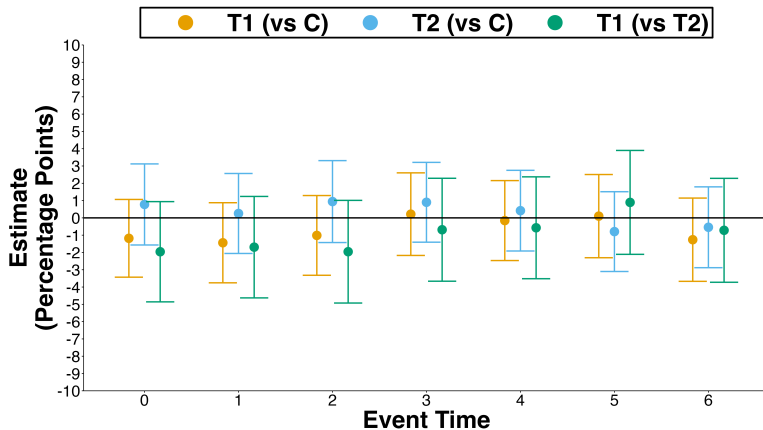


Online Experiment Example (Guttman-Kenney, 2024)



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Effects On Primary Outcome: Payments (% Statement Balance)



δ_6^k (percentage points, C mean: 27.8%):

T1 (vs C) -1.26 (s.e. 1.23) **T2 (vs C)** -0.54 (s.e. 1.19) **T1 (vs T2)** -0.72 (s.e. 1.53)

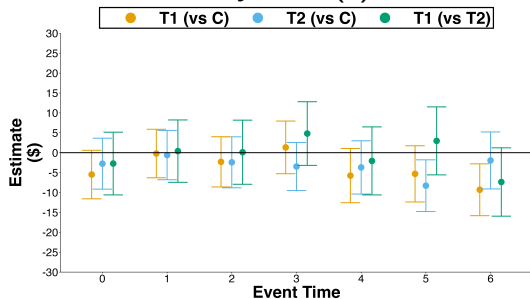
Effects On Payments (% Statement Balance) Pooled Across Cycles 0 to 6

Comparison	Pooled (δ^k)			
	(1)	(2)	(3)	(4)
T1 (vs C)	0.49 (1.06)	0.55 (1.06)	-0.67 (0.81)	-2.39 (1.23)
T2 (vs C)	0.35 (1.05)	0.37 (1.05)	0.29 (0.80)	1.04 (1.16)
T1 (vs T2)	0.14 (1.35)	0.18 (1.35)	-0.96 (1.02)	-3.44* (1.51)
Time F.E.		X	X	X
Consumer Controls			X	
Consumer F.E.				X

(percentage points, C mean: 34.6%)

No Significant $\uparrow$ Payments (\$) Or $\downarrow$ Revolving Debt (\$)

A. Payments (\$)



δ_6^k (percentage points):

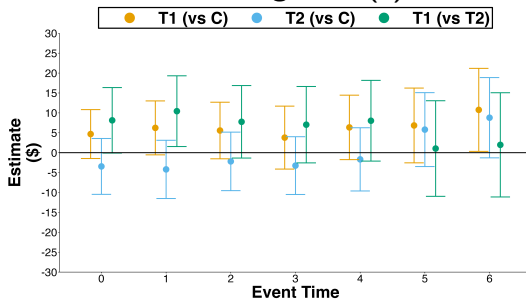
T1 (vs C) $-\$9.41^*$ (s.e. \$3.32)

T2 (vs C) $-\$1.95$ (s.e. \$3.65)

T1 (vs T2) $-\$7.36$ (s.e. \$4.38)

(C mean: \$58.4)

B. Revolving Debt (\$)



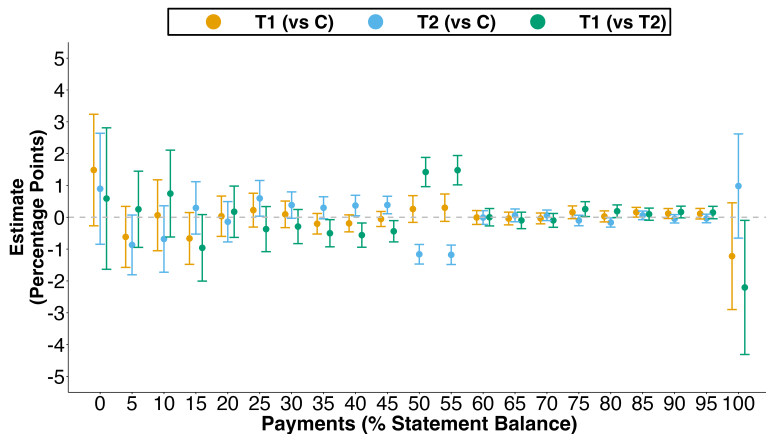
T1 (vs C): $\$10.76^*$ (s.e. \$5.32)

T2 (vs C): $\$8.79$ (s.e. \$5.14)

T1 (vs T2): $\$1.97$ (s.e. \$6.68)

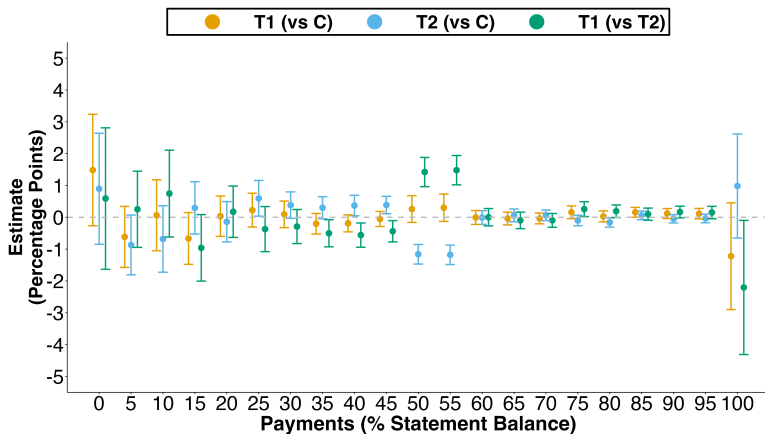
(C mean: \$230.9)

Limited Effects On The Distribution Of Payments (% Statement Balance)



T1 (vs T2): $\uparrow Pr(\text{Payment} \in \{(45\%, 55\%]\})$, $\downarrow Pr(\text{Payment} \in \{(30\%, 45\%], 100\%\})$

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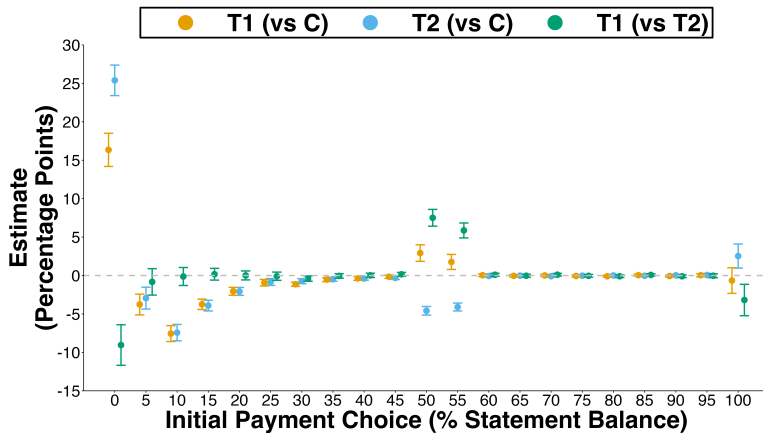


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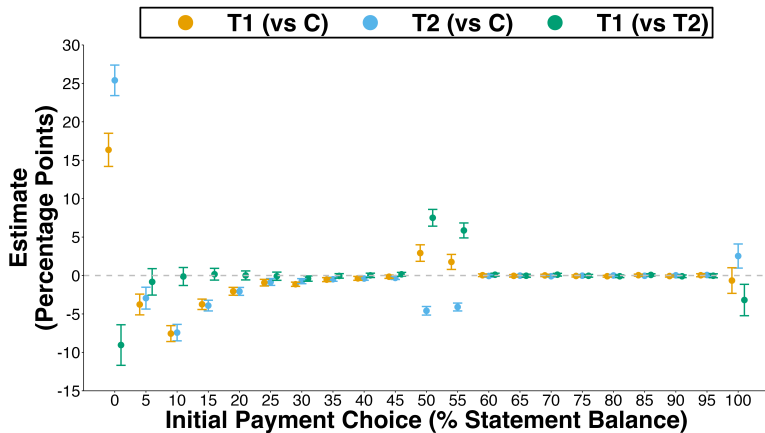
Shrouding minimum makes consumers more likely to pay small, round-numbered amounts
& less likely to pay multiples of the minimum payment

3. Mechanisms

Clickstream Data: Shrouding Minimum Lowers Initial Payment Choice



Clickstream Data: Shrouding Minimum Lowers Initial Payment Choice



Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) **T2 (vs C)** 30.50*** (s.e. 1.02)

T1 (vs T2) 11.06*** (s.e. 1.38) (percentage points, C mean: 33.7%)

Cardholders Not At Borrowing Constraints But Frequently Lack Liquid Cash

1. **Credit Card Portfolio Utilization (Equifax):** 25% (mean) & 36% (median)
 - Only 16% cardholders with 90%+ portfolio utilization
 - Typically not at borrowing constraint (inconsistent with liquidity constraints in HANK)
 - No significant heterogeneous effects for low-utilization consumers

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2. **Liquid Cash (Plaid):** *Minimum* liquid cash balances in 90 days pre-experiment
 - 63% of cardholders have <\$1, 93% <\$501, 96% <\$1,001 minimum liquid cash balances
 - No significant heterogeneous effects by minimum liquid cash balances, however, cannot rule out meaningful positive/negative effects for \$1,001+ group ($N = 292$)
 - Similar pattern for large U.K. bank credit cardholders (Guttman-Kenney et al., 2025)

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My results support macroeconomic modeling approach of Lee & Maxted (2026), in which present bias jointly explains high MPCs & revolving modest credit card debt (Meier & Sprenger, 2010; Kuchler & Pagel, 2021; Laibson et al., 2026)

Understanding Credit Card Behaviors & Reconciling Literature

Bunching of credit card payments at/just-above minimum payment amount makes sense given:

- Cardholders frequently have low liquid cash balances
- Discontinuity in economic payoffs from paying at least minimum e.g., to avoid late fees

Instead of being an irrelevant number to anchor to, the **minimum payment amount appears to be a relevant target** for consumers to meet/exceed (Bartels, Herzog, & Sussman, 2024)

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Reference-dependent preferences: Targeting to pay at least the minimum, and, if possible, pay their balance in full, can also help to understand other results in the credit card literature:

- Suggesting intermediate payment amounts, providing information disclosure or informational nudges on payment time/costs does not change debt
(Agarwal et al., 2015; Keys & Wang, 2016; Seira et al., 2017; Adams et al., 2022)
- Increasing minimum payments reduces debt of unconstrained cardholders
(Keys & Wang, 2019; Medina & Negrin, 2022; Castellanos et al., 2026; Allen et al., 2026)

Key Findings

1. No increase in payments from shrouding the minimum payment
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3. 50% payment option also does not change payments

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Understand why low credit score credit cardholders

often only pay at or just-above the minimum payment amount:

- Cardholders frequently have low liquid cash balances but not at borrowing constraint
- Minimum payment amount is a (relevant) target **not** an (irrelevant) anchor

Thank you!



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Contribution Relative to Guttman-Kenney, Adams, Laibson, Stewart, & Leary (2025)

Results are consistent, despite different treatments and populations

1. Shroud minimum payment in manual payment vs. shrouding autopay minimum at card opening
2. Even without offsetting responses found in my earlier paper (autopay enrollment, offsetting higher autopay with lower manual payments), shrouding remains ineffective
3. Test a new intermediate payment option: 50% of the statement balance
4. **Most importantly, the data and research design allow me to pin down the economics and psychology behind credit card payment decisions**

Pre-Experiment Controls

Linear controls:

- Age
- Credit Score
- Card Tenure
- Pre-experiment values of all of the primary and secondary outcomes
- Number of open credit cards held in their portfolio
- Total credit card portfolio balances
- Total credit card portfolio credit limits

Fixed effects:

- Interest rate
- Credit limit
- Autopay type

Primary and Secondary Outcomes

Primary Outcome: 1. Payments (%) = $\sum \text{Payments} / \text{Statement Balance}$)

Secondary Outcomes:

2. Payments (\$)
3. Statement Balance (\$)
4. Spending (\$)
5. Spending (%)
6. Revolving Debt (\$)
7. Credit Limit (\$)
8. Miss Pay (%)
9. Pay Min (%)
10. Pay 50% (%)
11. Pay Full (%)
12. Pay Current Balance (%)

Attrition (3, 6 Cycles)

	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
3	-0.23 (0.14)	-0.12 (0.18)	-0.12 (0.18)	0.37%
6	-0.09 (0.64)	0.00 (0.70)	-0.09 (0.83)	5.00%
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Pre-Experiment Summary Statistics

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)
Age	45.51	45.68	45.84	0.17 (0.38) [0.652]	0.33 (0.39) [0.390]
Credit Score	561.70	561.06	562.10	-0.64 (2.06) [0.757]	0.40 (2.10) [0.847]
APR (%)	15.71	15.72	15.72	0 (0.01) [0.826]	0 (0.01) [0.850]
Tenure (days)	265.58	257.53	262.24	-8.05 (5.10) [0.114]	-3.34 (5.10) [0.512]
Autopayer (%)	41.19	43.59	39.69	2.39 (1.68) [0.154]	-1.50 (1.65) [0.362]
Payments (%)	45.06	48.00	44.34	2.95 (1.44) [0.040]	-0.72 (1.39) [0.608]
Payments	77.07	85.00	76.99	7.93 (4.21) [0.060]	-0.08 (3.79) [0.984]
Statement Balance	207.94	206.38	208.75	-1.55 (7.57) [0.837]	0.82 (7.41) [0.912]
Spending	98.13	99.81	101.30	1.68 (4.67) [0.719]	3.17 (4.82) [0.510]
Spending (%)	50.11	51.95	50.25	1.84 (1.46) [0.207]	0.14 (1.44) [0.925]
Revolving Debt	149.09	144.62	149.98	-4.47 (7.15) [0.532]	0.89 (6.98) [0.898]
Credit Limit	292.09	287.64	294.11	-4.45 (8.16) [0.585]	2.02 (8.13) [0.803]
Miss Pay (%)	17.00	14.59	15.88	-2.41 (1.21) [0.047]	-1.12 (1.24) [0.365]
Pay Min (%)	17.48	17.38	16.79	-0.10 (1.29) [0.937]	-0.69 (1.26) [0.583]
Pay 50% (%)	3.74	3.07	4.20	-0.68 (0.60) [0.257]	0.45 (0.67) [0.497]
Pay Full (%)	31.73	34.39	30.02	2.66 (1.61) [0.098]	-1.71 (1.55) [0.270]
Pay Current Balance (%)	9.25	11.15	9.67	1.91 (1.05) [0.070]	0.42 (0.99) [0.668]

T-Tests (3 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	32.92	34.40	33.83	1.48 (1.39) [0.285]	0.92 (1.37) [0.503]	0.57 (1.76) [0.747]
Payments	59.01	63.18	55.75	4.16 (4.00) [0.297]	-3.26 (3.61) [0.367]	7.42 (4.81) [0.123]
Statement Balance	231.01	233.85	225.52	2.84 (7.93) [0.720]	-5.49 (7.65) [0.473]	8.33 (9.89) [0.400]
Spending	71.94	73.64	67.50	1.70 (4.54) [0.708]	-4.43 (4.27) [0.299]	6.14 (5.57) [0.271]
Spending (%)	36.03	38.04	36.55	2.01 (1.43) [0.158]	0.52 (1.42) [0.714]	1.49 (1.82) [0.412]
Revolving Debt	187.21	187.88	182.58	0.67 (7.70) [0.931]	-4.63 (7.48) [0.536]	5.30 (9.65) [0.583]
Credit Limit	320.21	319.02	321.35	-1.20 (8.75) [0.891]	1.14 (8.64) [0.895]	-2.34 (11.02) [0.832]
Miss Pay (%)	28.82	29.58	29.09	0.76 (1.55) [0.622]	0.28 (1.53) [0.857]	0.49 (1.96) [0.803]
Pay Min (%)	16.20	11.63	11.62	-4.57*** (1.12) [0.000]	-4.58*** (1.11) [0.000]	0.01 (1.38) [0.995]
Pay 50% (%)	2.72	3.35	0	0.63 (0.60) [0.293]	-2.72*** (0.24) [0.000]	3.35*** (0.55) [0.000]
Pay Full (%)	22.76	24.19	24.25	1.42 (1.45) [0.326]	1.48 (1.44) [0.303]	-0.06 (1.84) [0.974]
Pay Current Balance (%)	5.90	5.86	6.50	-0.04 (0.80) [0.960]	0.60 (0.82) [0.470]	-0.64 (1.03) [0.539]

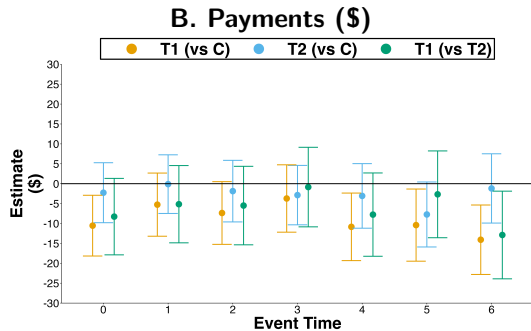
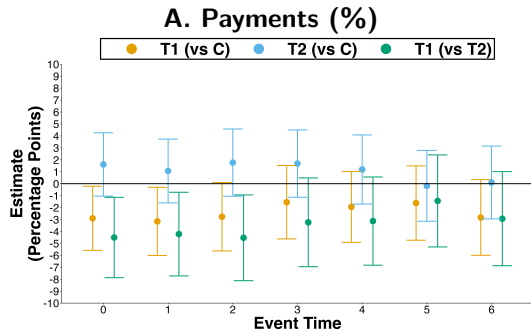
T-Tests (6 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	27.75	27.52	27.19	-0.23 (1.31) [0.864]	-0.56 (1.29) [0.664]	0.33 (1.65) [0.839]
Payments	58.41	51.53	57.07	-6.89 (3.62) [0.057]	-1.34 (3.95) [0.734]	-5.55 (4.73) [0.241]
Statement Balance	278.41	282.85	283.45	4.44 (8.81) [0.614]	5.04 (8.52) [0.554]	-0.60 (11.05) [0.957]
Spending	65.25	62.05	62.15	-3.20 (4.59) [0.486]	-3.10 (4.28) [0.468]	-0.10 (5.59) [0.986]
Spending (%)	27.32	28.59	27.99	1.27 (1.34) [0.346]	0.66 (1.32) [0.616]	0.61 (1.70) [0.721]
Revolving Debt	230.91	239.63	240.26	8.72 (8.59) [0.310]	9.35 (8.25) [0.257]	-0.63 (10.74) [0.953]
Credit Limit	366.11	362.14	369.84	-3.97 (9.86) [0.687]	3.73 (9.76) [0.702]	-7.70 (12.47) [0.537]
Miss Pay (%)	35.97	39.83	37.91	3.86* (1.69) [0.023]	1.94 (1.67) [0.246]	1.92 (2.14) [0.371]
Pay Min (%)	15.62	10.56	10.46	-5.06*** (1.11) [0.000]	-5.16*** (1.10) [0.000]	0.10 (1.35) [0.940]
Pay 50% (%)	1.85	2.71	0.10	0.86 (0.55) [0.116]	-1.76*** (0.23) [0.000]	2.62*** (0.52) [0.000]
Pay Full (%)	18.24	16.96	17.37	-1.28 (1.31) [0.328]	-0.87 (1.31) [0.509]	-0.41 (1.66) [0.803]
Pay Current Balance (%)	4.06	3.78	3.74	-0.28 (0.67) [0.678]	-0.31 (0.66) [0.636]	0.04 (0.84) [0.965]

Dynamic Effects on Payments (% Statement Balance) After 3 & 6 Cycles

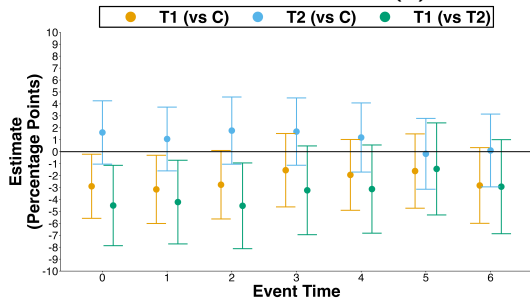
Comparison	Dynamic (δ_3^k)					Dynamic (δ_6^k)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 (vs C)	-0.21 (1.33)	1.47 (1.39)	0.22 (1.22)	-1.55 (1.57)	-7.09*** (1.26)	0.01 (1.30)	-1.26 (1.23)	-2.83 (1.62)
T2 (vs C)	-0.78 (1.31)	1.00 (1.37)	0.90 (1.18)	1.68 (1.44)	-7.42*** (1.24)	-0.38 (1.29)	-0.54 (1.19)	0.10 (1.56)
T1 (vs T2)	0.57 (1.75)	0.47 (1.75)	-0.69 (1.52)	-3.23 (1.90)	0.33 (1.64)	0.40 (1.64)	-0.72 (1.53)	-2.93 (2.01)
Time F.E.		X	X	X		X	X	X
Consumer Controls			X				X	
Consumer F.E.				X				X

With Consumer F.E.: Payments

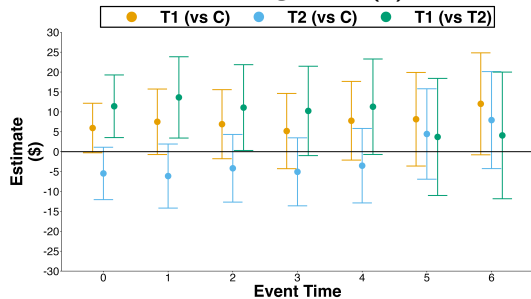


With Consumer F.E.: Statement Balance (\$) & Revolving Debt (\$) 🚀

A. Statement Balance (\$)



B. Revolving Debt (\$)



Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Payments	1.33 (3.37)	-3.48 (3.08)	4.81 (4.08)
Statement Balance	5.10 (3.65)	-4.48 (3.38)	9.57* (4.48)
Spending	-2.69 (3.68)	-4.57 (3.60)	1.88 (4.59)
Spending (%)	0.09 (1.20)	0.39 (1.19)	-0.31 (1.51)
Revolving Debt	3.79 (4.03)	-3.25 (3.70)	7.03 (4.90)
Credit Limit	0.39 (2.20)	-0.50 (2.05)	0.89 (2.75)
Miss Pay (%)	1.42 (1.47)	0.45 (1.44)	0.98 (1.85)
Pay Min (%)	-4.17*** (1.04)	-3.93*** (1.06)	-0.24 (1.3)
Pay 50% (%)	0.71 (0.60)	-2.88*** (0.26)	3.59*** (0.56)
Pay Full (%)	0.02 (1.31)	1.66 (1.26)	-1.64 (1.63)
Pay Current Balance (%)	-0.32 (0.78)	0.60 (0.80)	-0.92 (1.01)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Robustness Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C)			T2 (vs C)			T1 (vs T2)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Payments	-0.07 (3.84)	4.03 (4.00)	-3.72 (4.32)	-7.50* (3.44)	-3.09 (3.61)	-2.87 (3.81)	7.42 (4.81)	7.12 (4.82)	-0.85 (5.10)
Statement Balance	-3.34 (7.84)	2.45 (7.89)	4.61 (4.45)	-11.67 (7.55)	-6.00 (7.63)	-5.67 (3.79)	8.33 (9.89)	8.45 (9.87)	10.28* (5.19)
Spending	1.07 (4.35)	1.07 (4.53)	-0.42 (4.80)	-5.07 (4.06)	-4.77 (4.27)	-8.11 (4.92)	6.14 (5.57)	5.84 (5.58)	7.69 (6.14)
Spending (%)	2.46 (1.37)	1.93 (1.43)	-0.01 (1.60)	0.97 (1.36)	0.42 (1.42)	0.26 (1.58)	1.49 (1.82)	1.51 (1.81)	-0.27 (2.02)
Revolving Debt	-1.39 (7.58)	0.23 (7.65)	5.18 (4.82)	-6.69 (7.36)	-5.17 (7.46)	-5.06 (4.36)	5.30 (9.65)	5.39 (9.62)	10.24 (5.73)
Credit Limit	-5.61 (8.72)	-2.84 (8.68)	2.18 (2.68)	-3.28 (8.61)	-0.26 (8.56)	-0.84 (2.45)	-2.34 (11.02)	-2.58 (10.94)	3.02 (3.31)
Miss Pay (%)	2.79 (1.46)	0.84 (1.53)	3.24 (1.78)	2.30 (1.44)	0.38 (1.51)	1.29 (1.69)	0.49 (1.96)	0.46 (1.94)	1.95 (2.20)
Pay Min (%)	-4.76*** (1.05)	-4.70*** (1.12)	-4.62*** (1.42)	-4.77*** (1.04)	-4.69*** (1.11)	-3.84** (1.46)	0.01 (1.38)	-0.01 (1.37)	-0.78 (1.80)
Pay 50% (%)	0.75 (0.56)	0.68 (0.60)	1.36 (0.8)	-2.59*** (0.12)	-2.74*** (0.25)	-3.23*** (0.71)	3.35*** (0.55)	3.43*** (0.56)	4.59*** (0.95)
Pay Full (%)	0.36 (1.39)	1.33 (1.45)	-1.45 (1.77)	0.42 (1.38)	1.57 (1.44)	3.22* (1.60)	-0.06 (1.84)	-0.24 (1.84)	-4.66* (2.13)
Pay Current Balance (%)	-0.65 (0.74)	-0.08 (0.80)	-1.83 (1.23)	-0.01 (0.77)	0.55 (0.83)	0.18 (1.19)	-0.64 (1.03)	-0.63 (1.04)	-2.01 (1.55)
Time F.E.		X	X		X	X		X	X
Consumer F.E.			X			X			X

Effects on Secondary Outcomes After 6 Cycles

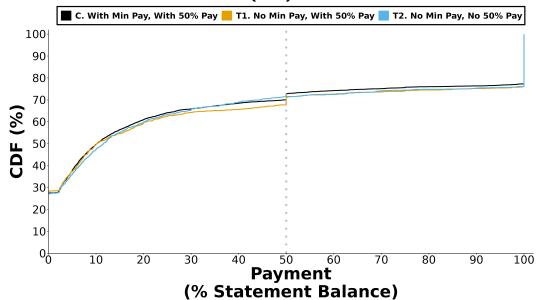
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-9.31** (3.32)	-14.08*** (4.45)	-1.95 (3.65)	-1.19 (4.44)	-7.36 (4.38)	-12.89* (5.62)
Statement Balance	6.03 (4.96)	5.54 (6.04)	3.92 (4.94)	3.77 (5.69)	2.11 (6.33)	1.78 (7.46)
Spending	-6.83 (4.27)	-4.28 (5.48)	-3.27 (3.99)	-6.73 (5.51)	-3.56 (5.27)	2.45 (7.00)
Spending (%)	-0.41 (1.28)	-0.26 (1.75)	0.71 (1.22)	0.52 (1.64)	-1.12 (1.6)	-0.78 (2.16)
Revolving Debt	10.76* (5.32)	12.02 (6.53)	8.79 (5.14)	7.94 (6.22)	1.97 (6.68)	4.09 (8.12)
Credit Limit	-2.15 (4.33)	-0.14 (4.97)	0.35 (4.17)	0.08 (4.74)	-2.50 (5.47)	-0.22 (6.22)
Miss Pay (%)	4.39** (1.63)	5.70*** (1.91)	1.92 (1.61)	2.71 (1.88)	2.47 (2.06)	2.99 (2.41)
Pay Min (%)	-4.60*** (1.09)	-4.82*** (1.56)	-4.34*** (1.06)	-3.99** (1.50)	-0.26 (1.32)	-0.84 (1.93)
Pay 50% (%)	0.94 (0.54)	1.60* (0.80)	-1.93*** (0.24)	-2.20*** (0.71)	2.87*** (0.52)	3.80*** (0.95)
Pay Full (%)	-2.44 (1.25)	-3.80* (1.78)	-0.72 (1.23)	0.66 (1.70)	-1.72 (1.57)	-4.46* (2.21)
Pay Current Balance (%)	-0.48 (0.66)	-1.96 (1.17)	-0.33 (0.66)	-0.79 (1.17)	-0.16 (0.84)	-1.18 (1.49)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Pooled Effects On Secondary Outcomes

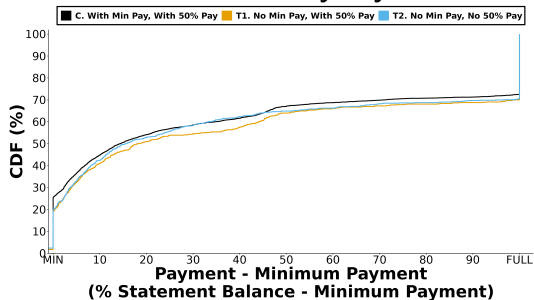
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-3.83 (2.19)	-8.85* (3.43)	-3.31 (2.13)	-2.73 (3.12)	-0.52 (2.74)	-6.12 (4.15)
Statement Balance	4.39 (2.42)	3.85 (3.63)	-2.29 (2.35)	-3.40 (3.08)	6.68 (3.04)	7.25 (4.21)
Spending	-2.49 (2.13)	-0.25 (3.93)	-2.32 (2.10)	-5.87 (4.11)	-0.17 (2.70)	5.61 (5.12)
Spending (%)	-0.38 (0.71)	-0.46 (1.29)	0.72 (0.71)	0.56 (1.25)	-1.10 (0.9)	-1.02 (1.61)
Revolving Debt	6.29* (2.78)	7.61 (3.95)	-0.10 (2.72)	-1.81 (3.64)	6.39 (3.50)	9.42* (4.77)
Credit Limit	-0.89 (1.72)	0.91 (2.26)	-0.09 (1.64)	-0.41 (2.11)	-0.80 (2.17)	1.32 (2.80)
Miss Pay (%)	1.72 (0.90)	3.43* (1.34)	1.29 (0.90)	2.18 (1.30)	0.43 (1.15)	1.26 (1.66)
Pay Min (%)	-3.53*** (0.65)	-3.94*** (1.17)	-3.38*** (0.64)	-3.29** (1.18)	-0.15 (0.80)	-0.65 (1.49)
Pay 50% (%)	0.70* (0.29)	1.35* (0.62)	-2.46*** (0.15)	-2.80*** (0.67)	3.16*** (0.28)	4.15*** (0.81)
Pay Full (%)	-1.00 (0.85)	-2.43 (1.42)	1.20 (0.82)	2.75* (1.30)	-2.20* (1.07)	-5.18*** (1.72)
Pay Current Balance (%)	-0.67 (0.40)	-2.17* (1.01)	0.26 (0.42)	-0.16 (0.97)	-0.93 (0.52)	-2.01 (1.26)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Distribution of Payments, After Three Cycles 🚀

A. CDF Payments (%)

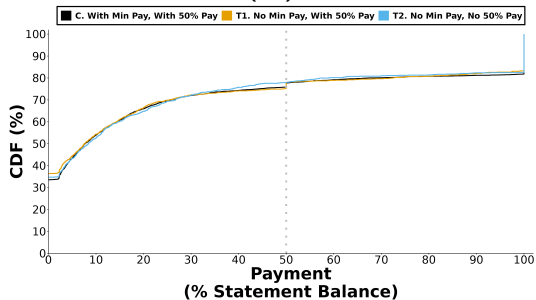


B. CDF Excess Payments (\$), Conditional On Any Payment

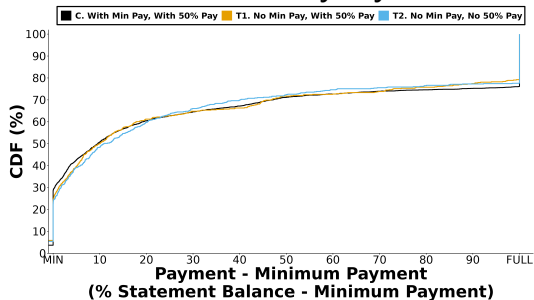


Distribution of Payments, After Six Cycles

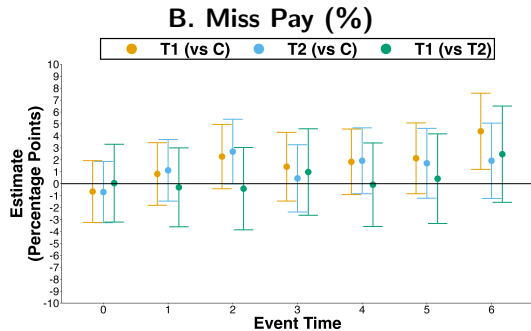
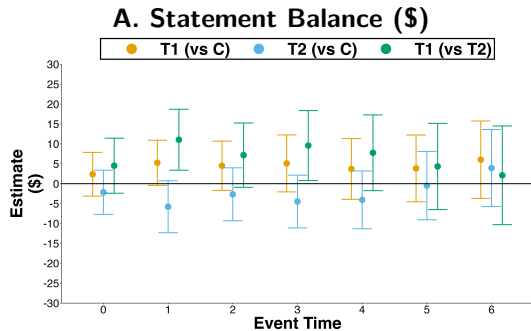
A. CDF Payments (%)



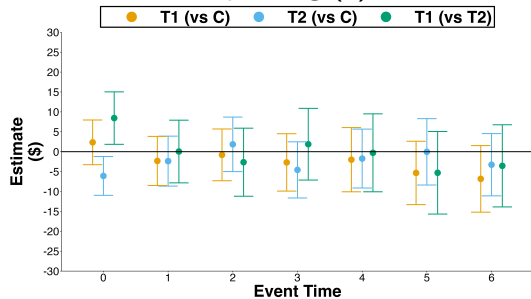
B. CDF Excess Payments (\$), Conditional On Any Payment



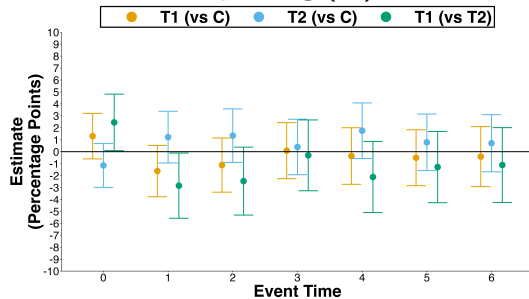
Statement Balance (\$), Miss Pay (%) 🚀



A. Spending (\$)

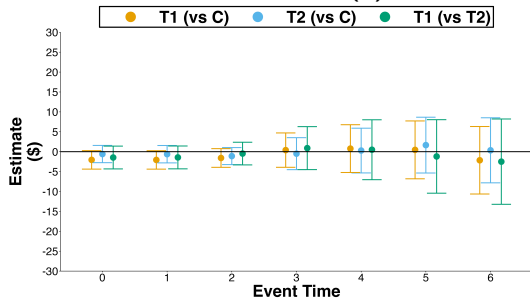


B. Spending (%)

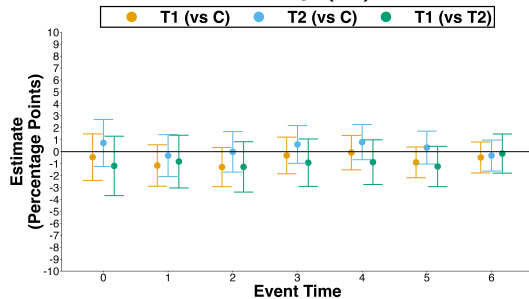


Credit Limit (\$) and Pay Current Balance (%) 🚀

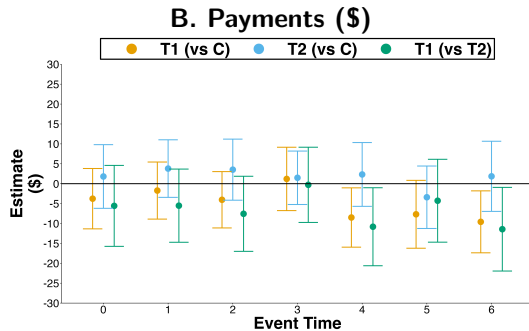
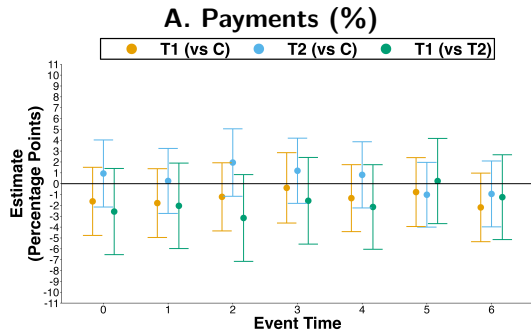
A. Credit Limit (\$)



B. Miss Pay (%)



No Autopay Subsample: Payments 🚀



No Autopay Subsample: Dynamic Effects After 6 Cycles

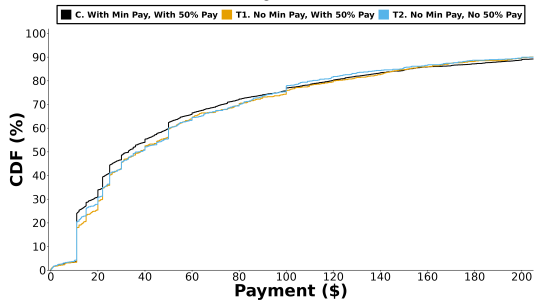
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments (%)	-2.18 (1.61)	-2.97 (2.08)	-0.94 (1.55)	-0.24 (2.02)	-1.24 (1.99)	-2.73 (2.58)
Payments	-9.58* (3.97)	-10.48* (5.08)	1.85 (4.50)	3.84 (5.37)	-11.44* (5.36)	-14.31* (6.51)
Statement Balance	5.66 (5.8)	3.30 (6.93)	8.07 (5.75)	8.87 (6.78)	-2.41 (7.42)	-5.57 (8.82)
Spending	-9.12 (5.10)	-7.76 (6.94)	1.15 (5.00)	0.10 (6.69)	-10.27 (6.48)	-7.86 (8.65)
Spending (%)	-1.12 (1.73)	-0.42 (2.37)	0.03 (1.58)	-0.39 (2.15)	-1.15 (2.11)	-0.03 (2.89)
Revolving Debt	10.31 (6.09)	9.85 (7.46)	9.13 (6.04)	7.11 (7.43)	1.18 (7.76)	2.74 (9.47)
Credit Limit	-3.17 (5.2)	-2.24 (5.78)	5.93 (5.20)	6.46 (5.89)	-9.09 (6.73)	-8.70 (7.50)
Miss Pay (%)	5.03* (2.21)	5.38* (2.55)	2.01 (2.12)	2.77 (2.42)	3.02 (2.75)	2.61 (3.16)
Pay Min (%)	-4.99*** (1.24)	-5.17*** (1.93)	-5.63*** (1.18)	-6.05*** (1.77)	0.64 (1.46)	0.88 (2.33)
Pay 50% (%)	0.81 (0.81)	2.22 (1.15)	-2.66*** (0.38)	-3.05*** (1.08)	3.47*** (0.78)	5.26*** (1.41)
Pay Full (%)	-3.84* (1.62)	-3.81 (2.35)	-1.33 (1.58)	0.43 (2.22)	-2.51 (2.01)	-4.24 (2.89)
Pay Current Balance (%)	-0.29 (0.92)	-0.96 (1.53)	-0.45 (0.88)	-0.42 (1.52)	0.16 (1.13)	-0.54 (1.94)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Effects On Portfolio of Credit Card Statement Balances 📌

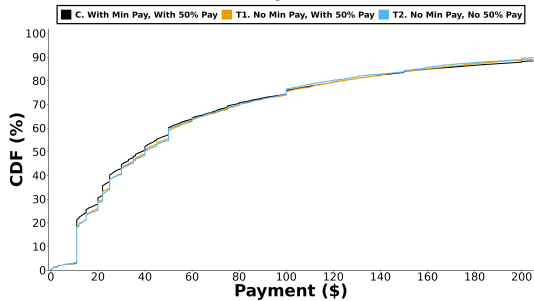
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Portfolio Statement Balances	36.55 (51.85)	6.60 (53.41)	29.94 (66.98)	\$3,681
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Distribution of Payments (\$) 🚀

A. Cycle 2



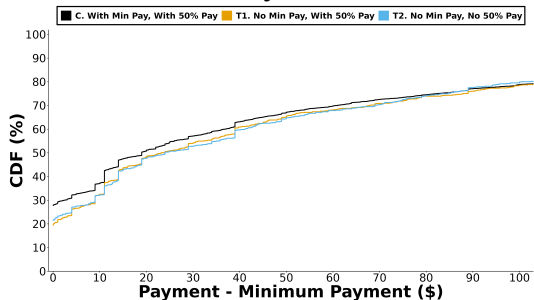
B. Pooled Cycles 0 to 6



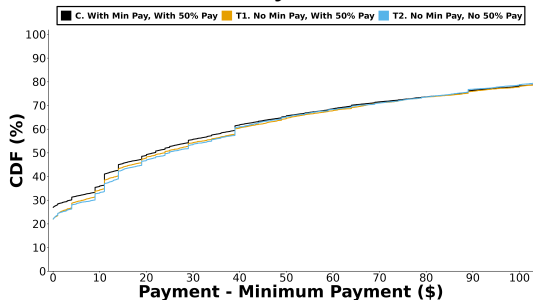
Distribution of Excess Payments (\$)



A. Cycle 2

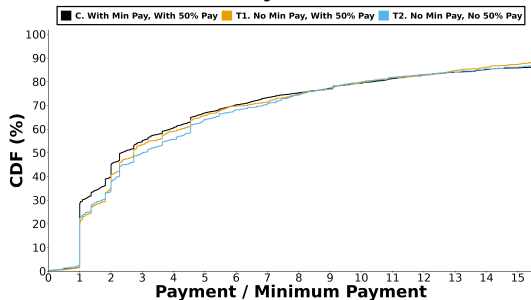


B. Pooled Cycles 0 to 6

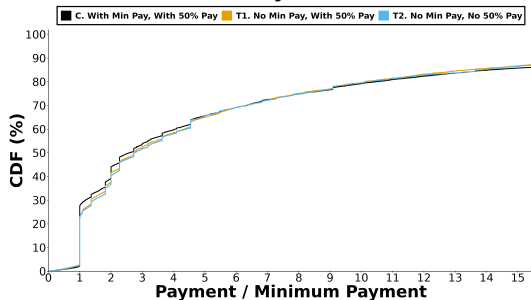


Distribution of Ratio of Payments To Minimum Payment

A. Cycle 2



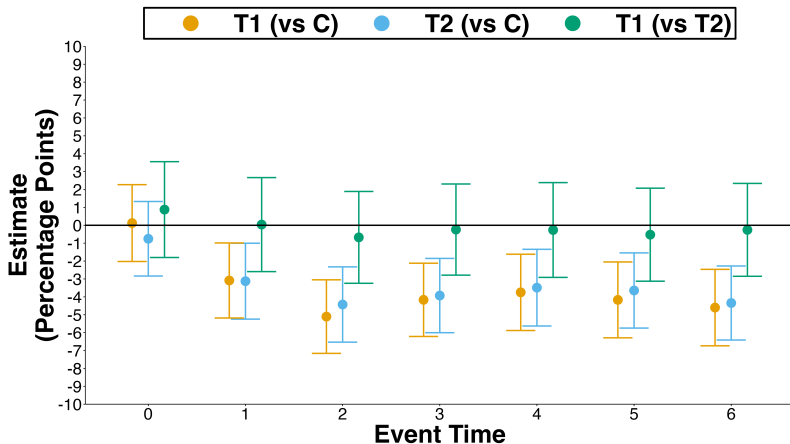
B. Pooled Cycles 0 to 6



Effects On Heuristic Payment Amounts

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Pay Multiples Of Minimum	−0.76*** (0.25)	−0.94*** (0.23)	0.18 (0.29)	3.48%
Pay Round Number	2.51*** (0.62)	3.61*** (0.62)	−1.10 (0.80)	9.58%
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

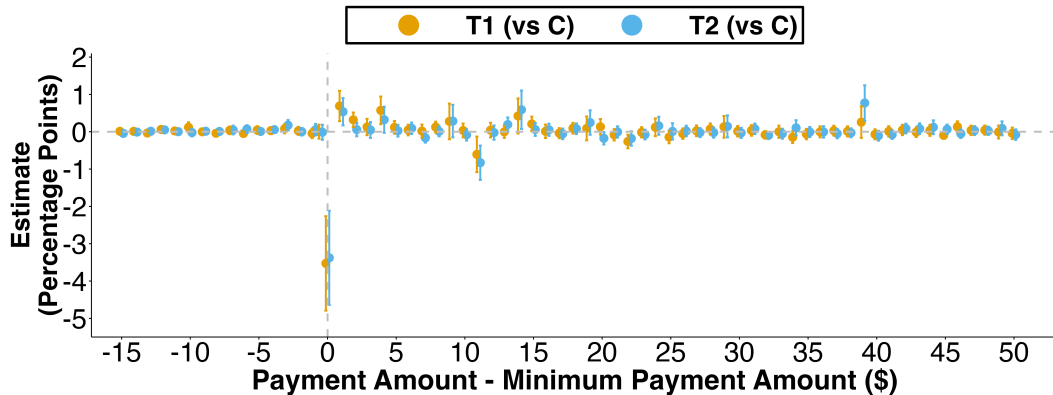
Effects On Payment Distribution: 29% Less Likely To Only Pay The Minimum



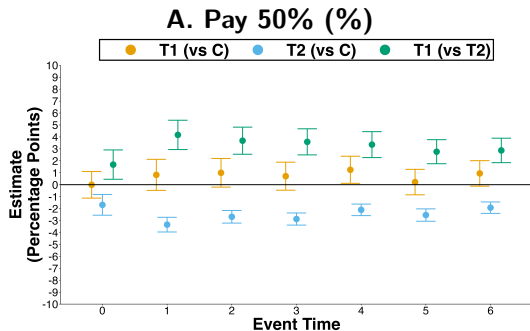
δ_6^k (percentage points, C mean: 15.6%):

T1 -4.60^{***} (s.e. 1.09) **T2** -4.34^{***} (s.e. 1.06) **T1 (vs T2)** -0.26 (s.e. 1.32)

Effects On Payments Proximate To Minimum Payment Amount

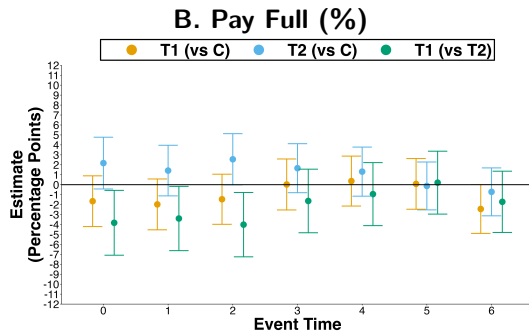


50% Payment Option Makes Consumers Initially Less Likely To Pay In Full



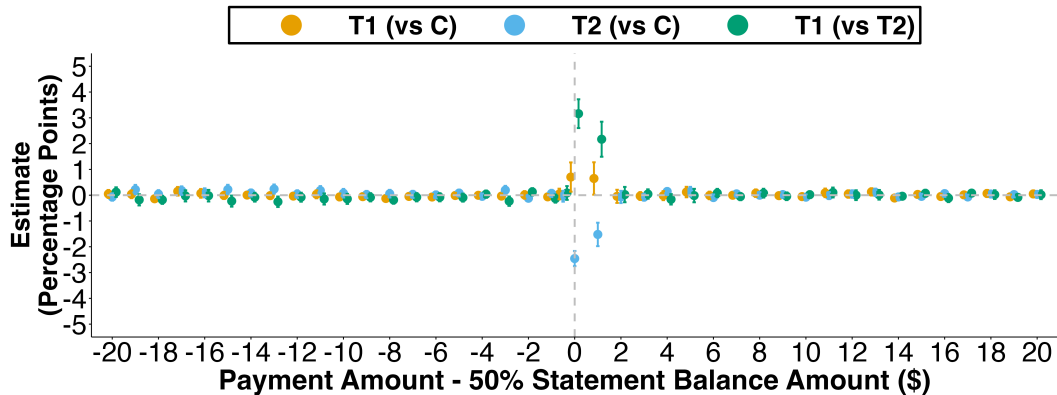
δ_6^k (percentage points):

T1 (vs C) 0.94 (s.e. 0.54)
T2 (vs C) -1.93*** (s.e. 0.24)
T1 (vs T2) 2.87*** (s.e. 0.52)
(C mean: 1.9%)

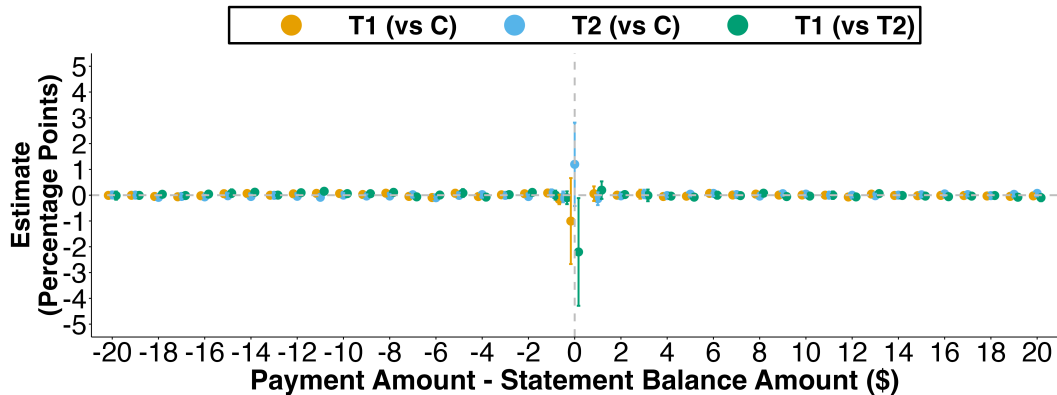


T1 (vs C) -2.44 (s.e. 1.25)
T2 (vs C) 0.72 (s.e. 1.23)
T1 (vs T2) -1.72 (s.e. 1.57)
(C mean: 18.2%)

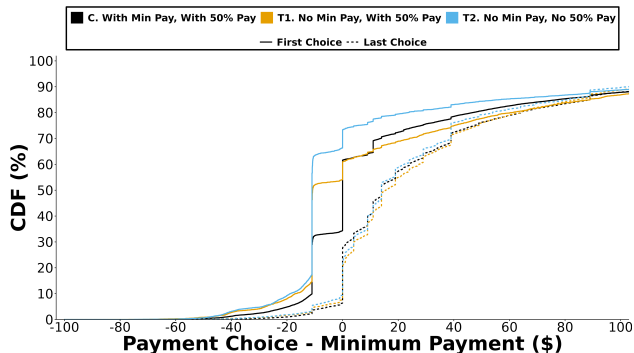
Effects On Payments Proximate To 50% Statement Balance Amount 🚀



Effects On Payments Proximate To Statement Balance Amount 🚀



CDFs Of Initial Vs. Last Payment Choices Within Payment Cycle



Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) **T2 (vs C)** 30.50*** (s.e. 1.02)

T1 (vs T2) 11.06*** (s.e. 1.38) (percentage points, C mean: 33.7%)

A. Initial Payment Choices

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Initial Payment Choice (% Balance)	-0.90 (0.87)	-4.33*** (0.81)	3.43*** (1.07)	26.83%
Pr (Initial Payment Choice < Minimum)	19.44*** (1.16)	30.50*** (1.02)	-11.06*** (1.38)	33.74%
Number of Payment Choices	0.40*** (0.11)	0.85*** (0.11)	-0.46*** (0.14)	3.82
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

A. Initial Payment Choices

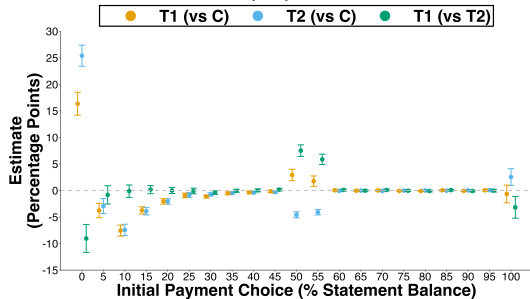
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Initial Payment Choice (% Balance)	-0.90 (0.87)	-4.33*** (0.81)	3.43*** (1.07)	26.83%
Pr (Initial Payment Choice < Minimum)	19.44*** (1.16)	30.50*** (1.02)	-11.06*** (1.38)	33.74%
Number of Payment Choices	0.40*** (0.11)	0.85*** (0.11)	-0.46*** (0.14)	3.82
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

B. Initial Payment Choices, Excluding Blanks & Zeros

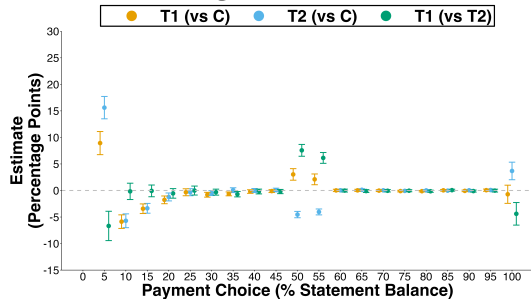
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Initial Payment Choice (% Balance)	-0.22 (0.86)	-1.82* (0.83)	1.60 (1.07)	30.19%
Pr (Initial Payment Choice < Minimum)	15.59*** (1.08)	23.57*** (1.04)	-7.98*** (1.38)	21.73%
Number of Payment Choices	0.10 (0.11)	0.35*** (0.11)	-0.25* (0.14)	3.01
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Effects On Initial Payment Choice 🚀

A. Initial Payment Choices (%)

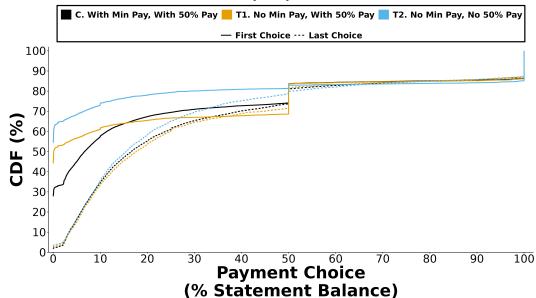


B. Initial Payment Choices (%), Excluding Blanks & Zeros

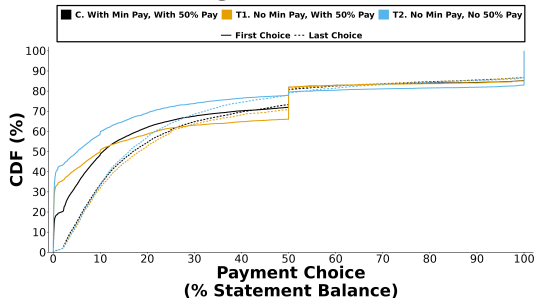


Initial Vs. Last Payment Choices Within Payment Cycle 🚀

A. CDF Initial Payment Choices (%)



B. CDF Initial Payment Choices (%), Excluding Blanks & Zeros



Empirical Support For New Macroeconomic Modeling Approach

- Heterogeneous agent models (“HANK”) include hand-to-mouth consumers with zero liquid wealth **experiencing binding borrowing constraints** (Kaplan & Violante, 2022)
- Lee & Maxted (2026): **binding borrowing constraints inconsistent with SCF data**
 - Consumers often borrow intermediate levels of credit card debt
 - This result could be driven by under-reporting of credit card debt in SCF (Zinman, 2009)
- I also find cardholders with low liquid cash are not borrowing constrained
- My empirical support for Lee and Maxted (2026)’s theory has broad implications
 - Affects how households respond to fiscal and monetary policy

Follow Keys & Wang (2019) Defining Near Minimum Payers 🚀

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

$$payments \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

Follow Keys & Wang (2019) Defining Near Minimum Payers 🗝️

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

$$\text{payments} \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Near-Minimum	2.89 (1.73)	-0.56 (1.54)	3.45 (2.07)	22.18%	1,226
Not-Near-Minimum	-1.44 (0.92)	0.28 (0.92)	-1.72 (1.17)	37.42%	5,488
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

Heterogeneous Effects By Pre-Experiment Credit Card Utilization

Portfolio utilization: 25% (mean), 36% (median)

16% cardholders with 90%+ portfolio utilization (borrowing constrained)

Heterogeneous Effects By Pre-Experiment Credit Card Utilization

Portfolio utilization: 25% (mean), 36% (median)
 16% cardholders with 90%+ portfolio utilization (borrowing constrained)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Q1 Card Utilization (< 60%)	-1.89 (1.65)	3.70* (1.60)	-5.59** (2.08)	52.23%	2,238
Q2 Card Utilization (60–91%)	1.57 (1.40)	-1.19 (1.31)	2.76 (1.72)	30.08%	2,238
Q3 Card Utilization (91%+)	-1.45 (1.15)	-1.30 (1.17)	-0.15 (1.44)	21.38%	2,238
Q1 Portfolio Utilization (< 12%)	-1.68 (1.43)	0.37 (1.47)	-2.05 (1.86)	33.24%	2,238
Q2 Portfolio Utilization (12–45%)	-2.31 (1.43)	0.94 (1.40)	-3.25 (1.83)	33.72%	2,238
Q3 Portfolio Utilization (45%+)	1.66 (1.45)	-0.96 (1.36)	2.62 (1.78)	36.85%	2,238
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

Heterogeneous Effects By Pre-Experiment Liquid Cash Balances

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Liquid Cash < \$1	1.32 (2.28)	1.01 (2.22)	0.31 (2.97)	25.15%	1,025
Liquid Cash \$1+	-1.36 (0.88)	0.20 (0.86)	-1.56 (1.10)	36.33%	5,689
Liquid Cash < \$501	-0.46 (1.06)	0.70 (1.06)	-1.16 (1.34)	31.37%	3,796
Liquid Cash \$501+	-0.80 (1.3)	-0.45 (1.23)	-0.35 (1.60)	38.78%	2,918
Liquid Cash < \$1,001	-1.11 (0.95)	0.42 (0.95)	-1.53 (1.21)	32.11%	4,726
Liquid Cash \$1,001+	0.25 (1.58)	-0.22 (1.50)	0.47 (1.96)	40.60%	1,988
Minimum Liquid Cash < \$1	-1.47 (1.02)	0.68 (1.03)	-2.15 (1.31)	32.10%	4,230
Minimum Liquid Cash \$1+	-0.25 (1.37)	-0.15 (1.33)	-0.09 (1.70)	38.99%	2,484
Minimum Liquid Cash < \$501	-0.88 (0.83)	0.61 (0.83)	-1.49 (1.06)	33.64%	6,252
Minimum Liquid Cash \$501+	1.50 (3.33)	0.43 (3.34)	1.07 (4.07)	48.04%	462
Minimum Liquid Cash < \$1,001	-0.95 (0.82)	0.21 (0.82)	-1.16 (1.04)	33.94%	6,422
Minimum Liquid Cash \$1,001+	1.77 (4.46)	5.87 (4.55)	-4.10 (5.77)	49.24%	292
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

Distinguishing Anchoring from Targeting

Anchors: *Irrelevant* numbers influence choices e.g., thinking of your (randomly-assigned) last two digits of SSN influences choices (Tversky & Kahneman, 1974; Ariely et al., 2003)

Targets (reference-dependent preferences): Consumers exert effort to meet/exceed *relevant* numbers e.g., marathon time goals (Heath, Larrick, & Wu, 1999; Allen et al., 2017)

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Distinguish using “**VPA**” **Value Proximity Analysis** (Bartels, Herzog, & Sussman, 2024)

VPA Predicts:

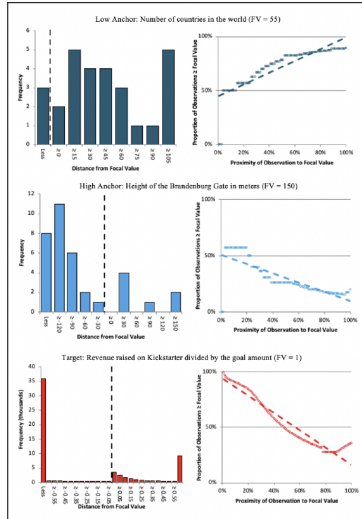
Anchoring: $\alpha \approx 50\%$ (symmetric), $\gamma > 0$ if low anchor ($\gamma < 0$ if high), $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} \neq 0$

Targeting: $\alpha \approx 100\%$ (asymmetric), $\gamma < 0$, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} = 0$

$$P(Y_w \geq F) = \alpha + \beta^{T1} T1_w + \beta^{T2} T2_w \\ + \gamma Proximity_w + \theta^{T1} (Proximity_w \times T1) + \theta^{T2} (Proximity_w \times T2) + \varepsilon_w$$

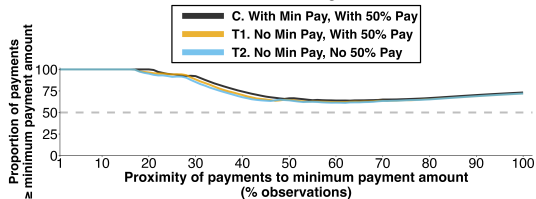
Example Histogram & VPA Charts (Bartels, Herzog, & Sussman, 2024) 🚀

Example Histograms and VPA Charts

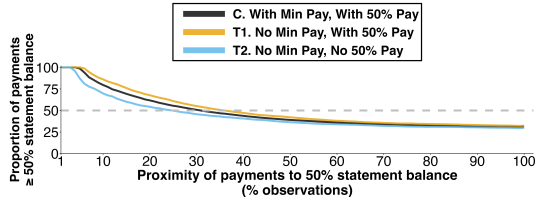


Value Proximity Analysis (VPA): Payments Proximate To Focal Amounts

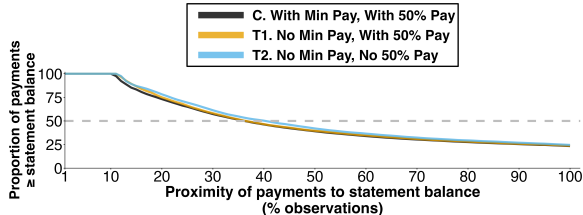
A. Minimum Payment



B. 50% Statement Balance



C. Statement Balance



VPA Results Inconsistent With Anchoring

	Minimum Payment	50% Statement Balance	Statement Balance
	(1)	(2)	(3)
Intercept	98.510*** (1.411)	76.831*** (2.718)	89.091*** (2.308)
T1	-1.403 (2.121)	5.290 (3.645)	1.380 (3.244)
T2	-2.840 (2.212)	-7.568 (3.986)	3.377 (3.052)
Proximity	-0.406*** (0.028)	-0.584*** (0.042)	-0.798*** (0.037)
Proximity $\times$ T1	-0.005 (0.041)	-0.044 (0.057)	-0.012 (0.052)
Proximity $\times$ T2	0.008 (0.042)	0.078 (0.062)	-0.016 (0.050)
Observations	300	300	300
R ²	0.647	0.773	0.873